DeepSight: Enhancing Deepfake Image Detection and Classification through Ensemble and Deep Learning Techniques

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Abstract:
in today's digital age, deepfake images poses a significant threat to multimedia content authenticity and integrity. Detecting and classifying deepfake images with high accuracy is crucial to addressing this growing challenge. Deepfake images, generated using advanced machine learning techniques, have become a significant concern due to their potential to deceive individuals and manipulate information. This paper presents DeepSight, an innovative approach that combines ensemble learning techniques with deep learning models to enhance deepfake image detection and classification. DeepSight leverages the diversity of multiple classifiers trained on distinct feature representations extracted from the input image data. The framework begins with an initial assessment to determine whether the image has been manipulated. Subsequently, the image goes through deep feature extraction using Convolutional Neural Networks (CNNs). The resulting feature vectors are then classified using Random Forest, KNearest Neighbors, and XGBoost, with hyper-parameter optimization. Experimental evaluations on benchmark datasets demonstrate DeepSight's superior performance compared to state-of-the-art methods, achieving higher accuracy and robustness across various deepfake generation techniques and quality levels. Notably, DeepSight achieves the highest accuracy of 97.5% when utilizing DenseNet and XGBoost. Overall, DeepSight represents a significant advancement in deepfake image detection and classification. It provides a reliable solution to address deceptive visual content in digital environments.

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SECTION I.
Introduction

As artificial intelligence advances, particularly deep learning, digital image creation and modification have completely changed. This technology has facilitated the creation of deepfakes, using complex algorithms to generate pictures and movies that are so realistic. This compromises the boundary between reality and fakery. According to Chesney and Citron [1], deepfakes pose a complex danger by degrading digital media reliability and affecting several areas such as politics, security, and personal rights. The growing trend towards virtual reality has prompted the urgent need for efficient detection and classification systems that differentiate between truth and synthetic deception.

However, identifying deepfakes remains challenging. It is becoming more and harder to tell deepfakes apart from real content, even with traditional detection methods and the unaided eye, due to the fast development of generative adversarial networks (GANs) and other machine learning techniques [2]. While deepfake technology is always developing and changing, current detection systems often depend on specific deep learning models. These models may not be applicable to the industry as a whole. It is crucial to have detection algorithms that can adjust to different degrees of complexity in deepfake movies and pictures and can withstand malicious attempts to defeat [3].

In light of these difficulties, studies investigating how deep learning and ensemble methods could improve deepfake detection have just started to appear Dolhansky et al., [4]. To improve detection rates and decrease false positives, ensemble approaches are considered. These methods combine the benefits of different learning algorithms. Similarly, deep learning techniques are being fine-tuned to identify deepfakes by analysing digital information for subtle characteristics. By presenting an innovative framework that uses the complementary strengths of ensemble and deep learning methods, this research hopes to contribute to the existing literature on deepfake detection. The aim is to develop a system that will better categorize deepfake pictures and reduce their dangers. The technologies that are currently implemented are unable to keep up with the rapid advancements that have been made to deepfake technology despite the significant progress that has been made in deepfake detection through deep learning. Many of these methodologies are unable to generalize across these diverse forms of synthetic media and can't defend themselves against sophisticated adversarial attacks. In order to enhance accuracy and minimise false positives in real-world settings, it is essential to develop adaptive and resilient ensemble approaches that incorporate various detection models.

A. Motivation

In our study motivated by the determination of combating the growing complexity and ubiquity of deep fake technologies. These technologies represent major dangers across society sectors. DeepSight is a framework for detecting deep fakes. These artificially created pictures and videos are becoming more indistinguishable from actual material, which leads to an increase in hazards associated with disinformation, political manipulation, and personal security Naitali et al., [5]. According to Le et al., [6], the fast breakthroughs in generative artificial intelligence have made the tools essential for making convincing deepfakes more accessible to the general public. This has resulted in an increased sense of urgency regarding strong detection systems. The requirement for a detection system that not only can adapt to deepfake's ever-evolving ways, but also retains high reliability and accuracy in a variety of circumstances, hence assuring the preservation of digital media integrity against these powerful threats, is the primary motivation behind our work Benazzouza et al. [7].

B. Taxonomy of the Deepfakes

The taxonomy of deepfakes classifies the many forms of this artificial media, which utilises advanced artificial intelligence to generate or modify digital information, often with a significant level of authenticity. The main categories consist of textual deepfakes, which create text that closely resembles human writing; deepfake videos, which overlay faces onto different bodies or modify facial expressions; deepfake images, which modify photographs to depict scenes or actions that did not happen; deepfake audio, which clones voices to produce fabricated speeches or conversations; and real-time deepfakes, which can manipulate live video feeds instantly. Each category utilises specialised technical techniques such as machine learning models and deep neural networks, which tackle certain obstacles and provide separate dangers about false information, privacy, and security.

Figure 1.
Taxonomy of Deepfakes

C. Contributions

The following are the contributions of our research work.

- Introduced DeepSight, a comprehensive framework that leverages ensemble machine learning, and deep learning techniques for effective detection and classification of deepfake images.
- Implemented a pre-processing stage that resizes images to fit CNN input requirements and incorporates an initial assessment to identify potential manipulations early in the process.

- Utilized advanced CNN architectures, including DenseNet, VGG16, and ResNet, for deep feature extraction crucial in detecting subtle discrepancies characteristic of deepfakes.
- Employed a various ensemble and ML classifiers such as Random Forest, K-Nearest Neighbors, and XGBoost, which were trained on distinct feature sets derived from image data to maximize detection and classification accuracy.
- Conducted rigorous performance evaluations on a publicly available dataset for deepfake image detection. Achieved a notably high accuracy rate of 97.5% with a combination of DenseNet and XGBoost, indicating a substantial advancement in the field.

D. Paper Structure

The paper's structure is outlined as follows: Section 2 reviews the literature relevant to the study, examining various works that have shaped the current understanding and methodology. Section 3 describes the study methodology, detailing the experimental design and data analysis approaches. Section 4 provides an in-depth discussion of the study's results, including a comparative analysis of state-of-the-art approaches. A conclusion is provided in Section 5, which summarizes the key findings and provides future research opportunities.

SECTION II.
Related Works

The evolving field of fake news detection has seen varied approaches leveraging advanced machine learning techniques, with studies increasingly focusing on enhancing model accuracy and handling complex data features. This section reviews critical contributions and identifies existing gaps, setting the stage for the novel methodologies introduced in our study.

Roy et al. [8] focused on developing deep learning models aimed at detecting and classifying fake news. Their efforts yielded promising results, achieving performance that was superior to existing methods in the field. However, a notable limitation of their research was the achieved overall accuracy rate of 44.87%. This accuracy, while a contribution to the domain, is relatively low when compared to more advanced models in the field of fake news detection. This highlights an area for potential improvement in future research efforts. A deep learning ensemble model developed by Aslam et al. [9] for detecting fake news was successfully trained using the LIAR dataset to achieve high accuracy rates. This model was designed to tackle misinformation proliferation on social media platforms effectively. In spite of its success, the study acknowledged certain limitations, including the lack of experience with machine learning for fake news detection and its complexities. The researchers suggested that further enhancements could be achieved by integrating additional features such as the speaker's job title and other relevant speaker information. The model could then be improved to identify fake news in the future.

Ali et al. [10 Introduced a deep ensemble model for fake news detection that leverages sequential deep learning techniques, showing superior detection accuracy on both the LIAR and ISOT datasets compared to existing models. However, the study primarily focuses on content-based features, which may limit its efficacy in analysing the brevity and nuance of news shared on social media platforms. This suggests a potential area for future research: a more comprehensive analysis of news origins and contextual factors. Moreover, the effectiveness of the model could be further enhanced by integrating learners that utilize features extracted through advanced embedding techniques. An ensemble learning model optimized by Self-Adaptive Harmony Search was used by Huang et al. [11] to detect fake news. This model demonstrated high accuracy rates and tackled some cross-domain intractability challenges. However, the study acknowledged that further enhancements are needed to more effectively address the issue of cross-domain intractability, indicating an area for potential improvement in future iterations of the model.

Javed et al. [12] developed an ensemble learning method to classify fake reviews, utilizing both textual and non-textual features, and achieved high F1 scores on the Yelp Filtered Dataset. Their approach includes a CNN-based architecture that processes n-gram embedding's to extract richer textual features through parallel convolutional blocks. However, the study faced limitations: the bag-of-words model used led to sparsity in text representation, and the neural networks demonstrated limited capabilities in handling unknown words. Furthermore, the integration of textual linguistic features with non-textual features related to reviewer behaviour presents complexities that could affect the model's performance across different contexts.

Kaliyar et al. [13] examined the application of machine learning ensemble methods for detecting fake news, focusing on different textual attributes to discern between real and fake news. Their research involved testing these methods across four real-world datasets, which demonstrated that ensemble learners significantly outperformed individual learners in terms of various performance metrics. Notably, the study does not identify any specific limitations or self-reported issues within the research, suggesting a confident application of the methods used and the results obtained. Using three widely-used machine learning models to classify news, Hakak et al. [14] achieved high training accuracy with their ensemble machine learning strategy for fake news detection. However, the study acknowledges several limitations: it builds on the foundations of previous studies that demonstrated poor accuracy, it may have suffered from a suboptimal selection of features, there were inefficiencies in parameter tuning, and the models were tested on imbalanced datasets, which could skew the results.

Research Gaps

Several gaps have been identified in the current scenario of fake news detection, and our research addresses them. Notably, studies such as Roy et al. [8] have highlighted issues with low accuracy rates, suggesting a need for more sophisticated models. Aslam et al. [9] and Ali et al. [10] have pointed out the limited use of contextual and content-based features, respectively, which could be enriched to enhance detection capabilities. Additionally, challenges like cross-domain intractability and the handling of complex and sparse textual data, as discussed by Huang et al. [11] and Javed et al. [12], respectively, indicate a need for models that better generalize across domains and utilize advanced text representation techniques to overcome linguistic hurdles. Moreover, the limitations noted by Hakak et al. [14] regarding suboptimal feature selection and the handling of imbalanced datasets underscore the necessity for improved model robustness and accuracy through enhanced methodological approaches. In order to achieve high accuracy and reliability in the area of fake news detection, we aim to integrate these insights through the development of a comprehensive model that integrates advanced machine learning techniques and robust feature integration.

SECTION III.
Proposed Methodology

The proposed methodology combines advanced ensemble and deep learning techniques to detect and classify deepfake images. Using multiple convolutional neural networks and diverse machine learning classifiers, we aim to significantly enhance deepfake detection accuracy and reliability. Figure 2 shows the overview of the proposed method. The technique is divided into three primary stages, as follows: 1. Image Pre-processing, 2. Deep Feature Extraction, and 3. Classification.

In the following subsections we will explain each phase in more detail.

- **Image Pre-processing**: This initial stage prepares the image for analysis by resizing it to meet the Convolutional Neural Network (CNN) specifications. Preprocessing also includes augmentation, gray scaling and pixel-level changes to detect anomalies that might indicate manipulation. This step is critical as it ensures that the images are in a uniform format and quality, allowing for more accurate analysis in subsequent stages.

- **Deep Feature Extraction**:

  In deepfake detection, advanced convolutional neural networks (CNNs) are used to extract features. The CNNs chosen are DenseNet, VGG16, and ResNet, based on their robustness in identifying complex patterns. Initially, images are pre-processed to standardize dimensions and normalize pixel values, ensuring uniform input for the CNNs. Each architecture processes the images to extract high-dimensional feature vectors that encapsulate essential attributes at varying levels of abstraction. Features from different CNNs may be pooled to reduce dimensionality and then concatenated to form a comprehensive feature set, combining the strengths of each network. These features are subsequently inputted into classifiers, such as Random Forest, K-Nearest Neighbors, and XGBoost with the entire model being trained and optimized on a labelled dataset using techniques like cross-validation to prevent overfitting. The final model is evaluated using validation sets to ensure effective generalization to new, unseen images, adjusting parameters based on performance metrics such as accuracy and precision. The combined approach facilitates an understanding of deep fake content from a nuanced perspective and allows for accurate classification.

- **DenseNet [15]**: DenseNet architectures establish dense connections between layers, promoting feature reuse and improving gradient flow, which can be beneficial for deepfake detection

- **VGG16 [16]**: An image classification task normally involves convolutional neural networks (CNNs), but these can also be used for deepfake detection thanks to their fine-tuned architectures.

- **ResNet [17]**: Various image classification benchmarks have shown that residual networks have high performance due to their depth. Models like ResNet-50, ResNet-101, and ResNet-152 can be used as feature extractors for deepfake detection.

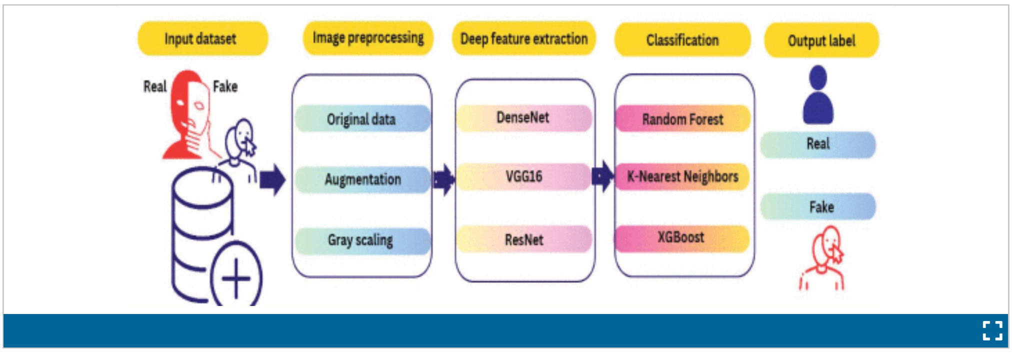

Figure 2.
Overview of the Proposed Methodology

Classification: Finally, using the features extracted in the previous step, sophisticated machine learning algorithms are employed to classify the images - Random Forest, K-Nearest Neighbors, and XGBoost [18].. This ensemble approach leverages multiple algorithms to improve detection accuracy and reliability. Each algorithm contributes differently to handling the data, which enhances the system's ability to discern genuine images from deep fakes effectively. This framework combines these steps together for detecting, classifying, and recognizing deep fake images with high precision. This framework addresses the challenges posed by sophisticated digital image manipulations. This methodology not only increases detection rates, but also minimizes false positives. Therefore, it is a valuable tool in efforts to combat digital misinformation.

SECTION IV.
Results and Discussion

The primary objective of this section is to analyze the deep learning models and ensemble machine learning algorithms. The evaluation has been broken down into several focused subsections to increase clarity and depth of analysis: **A. Performance Evaluation Metrics, B. Datasets** details the criteria used for model assessment, such as accuracy, precision, recall, and F1 score. It also describes the datasets that provide support for the research's applicability and validity. **C. Experimental Setup** outlines the technical specifications of the computing resources used, highlighting the rigorous computational framework necessary for executing the models' training and testing. **D. Performance Comparison and Discussion** offers a critical analysis comparing the performances of the models. It discusses each model's strengths and weaknesses and interprets these findings within the context of their potential real-world applications. This structured approach ensures a thorough examination of the models' capabilities, providing valuable insights into their practical implications and effectiveness.

a. Performance Analysis

For evaluating a model, a performance metric is necessary to determine its ability to predict outcomes. Here's an overview of commonly used metrics with their respective formulas:

Accuracy: An estimate of the proportion of true outcomes (both true positives and true negatives) in the data is given by this measure.

$$\text{Accuracy } = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

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The following are the meanings of TP, TN, FP, and FN: True Positives, True Negatives, False Positives, and False Negatives.

Precision: This metric evaluates the accuracy of a positive prediction, also called the positive predictive value.

$$\text{Precision } = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

View Source

Recall: This measures a model's ability to find all the relevant cases (i.e., True Positives) within a dataset.

$$Recall = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{3}$$

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F1-Score: As the F1-Score takes into account both false positives and false negatives, it is a harmonic mean of precision and recall.

Average-Precision (AP): Precision-recall curves can be summarized using AP, which weights precisions based on recall increases from previous thresholds:

$$\mathrm{AP} = \sum_n (R_n - R_{n-1}) P_n \tag{4}$$

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In this case, $\mathrm{P_n}$ and $\mathrm{R_n}$ represent precision and recall at the nth threshold, respectively.

In ROC analysis, the area under the curve (AUC-ROC) is used to estimate the performance of a classification across all classification thresholds. ROC curves plot true positive rate (Recall) against false positive rate (False Positive Rate) to determine the area under the curve.

$$\mathrm{AUC} - \mathrm{ROC} = \int_0^1 \mathrm{TPR}(x) dx \tag{5}$$

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Where, TPR is the true positive rate.

These metrics are essential for assessing different aspects of a model's predictions, including overall accuracy, error types, and the trade-offs between sensitivity and specificity. Each metric offers unique insights and is chosen based on the specific requirements and context of the modelling task.

b. Datasets

The dataset used in this study was compiled from Yonsei University's Computational Intelligence and Photography Lab and the renowned Celeb-DF dataset. This dataset offers a wide range of photographic and annotation materials to facilitate extensive study of artificial intelligence and photography. Several scenarios are provided for evaluating image processing algorithms. However, Celeb-DF uses top-notch, altered videos of celebrities to imitate authentic deepfakes. The analysis provides a solid basis for developing and evaluating deepfake detection tools. This study combines these datasets to gain advantages across a wide range of uses, ranging from scholarly research to practical applications in deepfake analysis. As a result, the resultant models are more reliable and have a greater extent. Incorporating this key dataset not only enhances the study but also allows for a thorough testing under a variety of actual and synthetic settings, which is crucial for improving the efficacy of the existing methods.

c. Experimental Setup

In this study, the experimental setup involved a computer equipped with four central processing units (CPUs), each featuring an i7 processor running at 2.5 GHz and equipped with 8 GB of RAM. This configuration provided the necessary computational power for the deep learning methodologies employed in the research. Python 3.7 was used to implement and evaluate the proposed model. It was chosen for its extensive libraries and wide support which facilitate robust data analysis and model testing. Table 1 and Table 2 details the classification performance of the models, including F1 score, recall, accuracy, and precision. Additionally, Figure 2 visually demonstrates the model's effectiveness in the initial binary classification step, showing how it effectively segmented data into preliminary categories. This setup highlights the practical application of the model and its capability to handle complex datasets. In addition to demonstrating the technical precision and analytical competence needed for such advanced research, it also highlights the importance of technical precision.

d. Performance Comparison

This section offers a critical analysis comparing the performances of the models. It discusses each model's strengths and weaknesses and interprets these findings within the context of their potential real-world applications. This structured approach ensures a thorough examination of the models' capabilities, providing valuable insights into their practical implications and effectiveness.

Figure 3 illustrates the analysis of the performance differences among three CNN deep learning models, namely VGG16, ResNet, and DenseNet, across a wide range of classification tasks highlights a number of notable differences. As the best performer, DenseNet achieves 97.54% accuracy, 96.23% precision, 95.63% recall, and 97.23% F1 score. The efficient feature utilization and deeper connectivity of DenseNet likely contribute to its superior ability to accurately identify relevant instances across diverse scenarios. Similarly, ResNet shows a robust performance with an accuracy of 94.75% and an F1 score of 93.89%, outperforming VGG16 with an accuracy of 89.43% and an F1 score of 87.58%. The architecture of DenseNet and ResNet makes feature extraction possible at a higher level than VGG16. As a result, DenseNet and ResNet can capture a more complex feature hierarchy as compared to VGG16, which has a simpler structure. Table 1 summarizes the performance comparison of the CNN models.

TABLE 1 Performance Comparison of the Cnn Models

Deep learning model performance				
CNN Models	**Accuracy**	**Precision**	**Recall**	**F1 score**
VGG16	89.43	86.48	84.59	87.58
ResNet	94.75	92.53	91.34	93.89
DenseNet	97.54	96.23	95.63	97.23

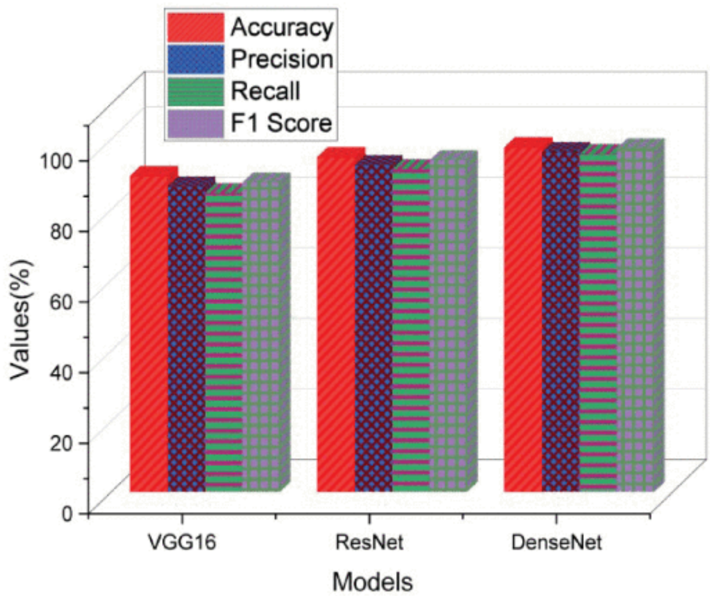

Figure 3.
DL Models' Performance Comparison

Figure 4 illustrates the performance comparisons between Random Forest, KNN, and XGBoost highlight differences in accuracy, precision, recall, and F1 score between the three popular ensemble learning models. XGBoost leads with impressive results, demonstrating the highest accuracy (97.59%), precision (95.47%), recall (94.31%), and F1 score (96.29%), making it exceptionally effective for complex classification tasks. In addition, KNN demonstrates strong performance, particularly in accuracy (93.6%) and F1 score (92.83%), showing its ability to handle diverse datasets. Despite trailing behind with an accuracy of 87.28%, Random

Forest still performs well with an F1 score of 85.75%. There are a number of applications for it, and it offers a high level of robustness and ease of use. As a result, XGBoost delivers the best results among the three models for handling complex data patterns as it utilizes advanced boosting techniques that effectively minimize errors sequentially. Table 2 summarizes the performance comparison of the ensemble-based models.

TABLE 2 Performance Comparison of the Ensemble Based Models

Ensemble based model performance				
Model	**Accuracy**	**Precision**	**Recall**	**F1 score**
Random Forest	87.28	85.68	83.77	85.75
KNN	93.65	91.67	89.79	92.83
XGBoost	97.59	95.47	94.31	96.29

Figure 5 illustrates the comparative analysis of three CNN deep learning models—VGG16, ResNet, and DenseNet—DenseNet stands out significantly across key performance metrics: accuracy, F1 score, and ROC-AUC. It demonstrated the highest scores in all three categories with an accuracy of 97.54%, an F1 score of 97.23%, and a ROC-AUC of 99. This indicates its exceptional capability to balance precision and recall effectively while also distinguishing between classes with high reliability. Following closely, ResNet shows strong performances with a 94.75% accuracy, 93.89% F1 score, and 94 ROC-AUC, making it a robust option for complex image recognition tasks. VGG16, while competent, offers lower performance figures with an 89.43% accuracy, 87.58% F1 score, and an ROC-AUC of 89. This analysis, potentially illustrated in Figure 5, illustrates that while VGG16 is reasonably capable, DenseNet and ResNet offer greater efficiency for applications demanding high precision and reliable classification. Table 3 summarizes the results of ROC-AUC, Accuracy and F1 score analysis of DL models. This marks them as preferable choices for more demanding computational tasks. Figure 6. Illustrates the sample real and fake images from the given dataset. Table 4 summarizes the comparative analysis of the state-of-the-art approaches and our proposed method.

TABLE 3 Roc-Auc, Accuracy and F1 Score Analysis of Dl Models

DL Model	**Accuracy**	**F1 score**	**ROC-AUC**
VGG16	89.43	87.58	89
ResNet	94.75	93.89	94
DenseNet	97.54	97.23	99

TABLE 4 Comparison with State-of-the-Art Approaches

Citations	**Method**	**Accuracy**
Lee et al. [19]	Shallow Fake Face Net	72.5%
Chandani et al.[20]	ResNet-152	76.7%
Ranjan et al [21]	CNN	95.8%
Qul ain et al[22]	VGG-16	92.09%
Wodajo et al[23]	CNN	91.5%
Mittal et al. [24]	AlexNet	55.8%
Proposed method	**DenseNet and XGBoost**	**97.5%**

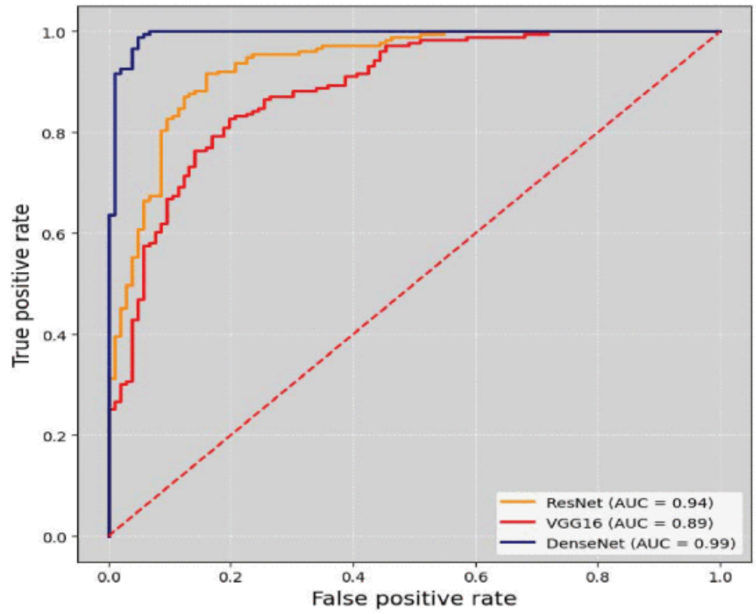

Figure 5.
ROC Curve Representation

Figure 6.
Sample Real and Fake Images from the Dataset

SECTION V.
Conclusion and Future Work

This study proposed DeepSight, a new framework that combines ensemble learning techniques with advanced deep learning models. This framework is designed to significantly enhance deepfake image detection and classification. By leveraging a variety of classifiers trained on unique feature representations and utilizing deep feature extraction through Convolutional Neural Networks (CNNs), DeepSight achieved a notable accuracy of 97.5% using DenseNet and XGBoost, compared to existing methods. This exceptional performance shows the effectiveness of integrating diverse analytical approaches to tackle advanced deepfake challenges. Moving forward, DeepSight development can be further advanced by incorporating additional Deep Learning (DL) architectures and more efficient ensemble techniques. This will improve accuracy and processing efficiency. There is also a significant opportunity to expand the diversity of the datasets used. This would allow the proposed system to address more complex and subtle deepfakes variations, enhancing its applicability across different digital environments. Moreover, incorporating DeepSight with real-time detection applications on digital media platforms could significantly mitigate the spread of deceptive content, thereby enhancing digital communications reliability and security across various sectors. These enhancements will ensure that DeepSight remains at the forefront of the fight against digital misinformation.